KLAi AI and Data Science Professional

24 sessions of 2 hours | 48 hours | free tools only

You arrive confident in Excel, with no code and no statistics. You leave able to take a business question from raw data to a tested model or analysis, explain it to a manager, and run it as a small app. Each session is concepts first, with laptops closed, then a lab where you work.

Module		Session title	What you keep from the lab	
A Think with data Sessions 1 to 4	1	**What data science is and how to think in it**	Notebook in Drive and your business question	
	2	**Data as tables: records, variables and keys**	Workbook dictionary tab: record, key, problems per tab	
	3	**Summarising and showing data fairly**	Workbook: three pivot tables and three charts	
	4	**Uncertainty, samples and cause**	Simulation tab: 200 samples of 30 orders	
B Python for data Sessions 5 to 9	5	**Your first Python: variables, lists and error messages, next to the spreadsheet**	Notebook: price list, one error read and fixed	
	6	**Decisions, repeats and functions: if, for, and a tested function**	Notebook: a tested loyalty_tier function	
	7	**pandas I: load, look and filter a table of 30,229 orders**	orders_profile.csv: a data profile	
	8	**pandas II: cleaning the mess, with a log and a reconciled total**	orders_clean.csv, cleaning log, reconciliation note	**Milestone 1**
	9	**pandas III: group, join and reshape**	Revenue tables and a join log with row counts	
C Analyse and explain Sessions 10 to 13	10	**SQL for analysts: ask a database the questions you ask pandas**	Query book: ten SQL queries with pandas proofs	
	11	**Exploring and visualising data with Python: from a new file to a finding**	Notebook: eight charts and three findings	
	12	**Inference: is the difference real, or could it be luck?**	Notebook: the offer test five ways, a decision note	
	13	**Regression: explaining and predicting a number**	regression_summary.csv and a one-page report	**Milestone 2**
D Machine learning Sessions 14 to 19	14	**Machine learning: the workflow, from loans data to a first model**	loans_prepared.csv and a three-sentence note	
	15	**Classification: predicting a yes or no answer with logistic regression and a decision tree**	loan_risk_scores.csv and a threshold note in UGX	
	16	**Judging a model: metrics, leakage and imbalance**	model_comparison.csv and a note on the leaked column	
	17	**Forests, boosting and overfitting: a stronger model without fooling ourselves**	Model comparison table and a decision memo	
	18	**Finding structure without labels: customer segments and unusual orders**	customer_segments.csv and unusual orders to review	
	19	**Forecasting and a first end-to-end model: baseline, time-based split and Pipeline**	Baseline, tested forecast, Pipeline, manager note	**Milestone 3**
E Neural networks and text Sessions 20 to 22	20	**Neural networks and deep learning, without the maths wall**	Notebook, results table and a recommendation	
	21	**Text as data and meaning as numbers**	feedback_scored.csv and a three-sentence finding	
	22	**Language models and building with them**	eval_results.csv and audit_log.csv	
F Ship it and present Sessions 23 to 24	23	**Ship it: reproducible work, a small app and responsible use**	Shipping folder: app.py, saved model, model card	
	24	**Present your course project and plan your path from here**	Presentation sheet and a six month learning path	**Project talk**

Every session

- 10 min recall and the question of the day
- 45 min concepts, laptops closed
- 10 min break
- 45 min lab: predict, run, read, change
- 10 min exit check, keep and next

Free tools, nothing to buy

- Google Colab and Google Sheets (or Excel)
- Python: pandas, NumPy, scikit-learn, matplotlib
- SciPy, statsmodels, seaborn, SQLite
- GitHub, Streamlit Community Cloud
- TensorFlow Playground, sentence-transformers

At the end you can

- Clean a raw export and prove its total
- Query a database with SQL and test a difference
- Build and judge a model against a baseline
- Publish a small app and present to a manager

Course project. Question in 13, dataset in 14, tested model in 19, app in 23, talk in 24. No exams or marks. Finishing it earns a KLAi certificate of completion.